NCCLbpf: Verified, Composable Policy Execution for GPU Collective Communication

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ABSTRACT

NCCL is the de facto standard library for GPU collective communication in large-scale distributed training. Its plugin system allows operators to customize collective behavior, but plugins execute as unverified native code that can crash the training job with a single null-pointer dereference. Moreover, plugins share no state across extension points, and updating a plugin requires restarting the job. NCCL's CPU-side decision path is latency-tolerant—the GPU collective it configures takes microseconds to milliseconds—making eBPF's verification overhead negligible while enabling safety, composability, and live updates. We present NCCLbpf, which embeds a userspace eBPF runtime inside NCCL's plugin interface to provide (1) load-time verification that catches crash-inducing bugs before execution, (2) typed eBPF maps for cross-plugin state sharing, and (3) atomic policy hot-reload without call loss, all with zero NCCL modifications. Our evaluation on NCCL 2.29.7 shows 52–100 ns of CPU overhead per decision (<0.06% of collective latency), correct rejection of all unsafe programs, sub-microsecond policy swaps, and elimination of intermittent 2× AllGather throughput degradation, with zero measurable GPU-level overhead.

1 INTRODUCTION

Large-scale GPU training depends on collective communication performance. Frameworks such as Megatron-LM [15], DeepSpeed [12], and PyTorch FSDP [19] rely on NCCL [9] for all inter-GPU data movement. The choice of algorithm (ring vs. tree), protocol (LL vs. Simple), and channel count significantly affects collective latency [2, 3]. NCCL exposes a plugin system—tuner, profiler, net, and env plugins—that allows operators to customize collective behavior, and major cloud providers already ship custom tuner plugins [2].

However, every plugin executes as unverified native code in NCCL's address space: a null-pointer dereference crashes the training job, an infinite loop hangs the communicator, and an off-by-one error can silently corrupt internal state. The tuner and profiler are independent extension points with no shared state mechanism, precluding closed-loop control. Updating a plugin requires restarting the job, leaving operators no way to iterate on policies at runtime.

The key insight is that NCCL's policy hot path (`getCollInfo`) runs on the CPU and is *latency-tolerant*: the GPU collective it configures takes microseconds to milliseconds, so 50–100 ns of eBPF overhead is negligible. This contrasts with eBPF's traditional domain (packet processing and syscall filtering), where nanosecond budgets are tight. Building on this observation, NCCLbpf embeds bpftime [18], a userspace eBPF runtime, inside a standard NCCL plugin. Policy authors write restricted C compiled to BPF bytecode, verified at load time, and JIT-compiled to native code. The system loads via standard `NCCL_TUNER_PLUGIN` and `NCCL_PROFILER_PLUGIN` environment variables with zero modifications to NCCL. This paper makes four contributions:

(1) **NCCLbpf**, a system that brings verified eBPF policy execution to NCCL's plugin interface (§3–4).
(2) A map-based composability model that enables closed-loop profiler-to-tuner adaptation, a capability absent from NCCL's native plugin architecture (§3).
(3) Atomic policy hot-reload with 0.774 µs swap time and zero call loss across 400,000 invocations (§3).
(4) An evaluation on NCCL 2.29.7 showing 52–100 ns CPU overhead per decision (<0.06% of collective latency), end-to-end GPU overhead within measurement noise, and elimination of intermittent 2× AllGather throughput degradation (§5).

2 BACKGROUND AND MOTIVATION

NCCL plugin architecture. NCCL organizes extensibility around four plugin types loaded via `dlopen`: the *tuner* plugin's `getCollInfo` selects algorithm, protocol, and channel count per collective call; the *profiler* plugin receives timestamped event callbacks; the *net* plugin controls network connections; and the *env* plugin sets runtime parameters. These plugins are independent extension points with no cross-plugin communication mechanism and no verification of safety.

The safety gap. Native plugins execute with full process privileges. A null dereference produces an immediate `SIGSEGV`, terminating the training job. An off-by-one error can silently corrupt NCCL internal state, degrading performance without any visible error. An infinite loop hangs the communicator indefinitely. These are not hypothetical risks: NCCL's own inspector plugin has caused production training crashes due to use-after-free and deadlock bugs [10], and profiler plugins have triggered segfaults on multi-GPU nodes [11]. At scale, such failures are costly: Llama 3 pre-training on 16,384 GPUs experienced 466 job interruptions in 54 days [4], and analyses of production GPU clusters find that ∼30% of training jobs fail, with programming and configuration errors as the dominant cause [7]. The absence of safe hot-reload further compounds the risk: rolling out plugin updates across a fleet requires restarting every training job.

Why eBPF fits. Extended BPF (eBPF) [16] is an execution framework, originally implemented in the Linux kernel for packet filtering and tracing, that statically verifies programs for memory safety and bounded execution before JIT-compiling them to native code. It has since been adopted as a general safe-extension mechanism across storage [21], networking [8, 22], and memory

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management [17], with recent userspace runtimes [18] extending its reach beyond the kernel. Three properties make it well-suited to NCCL policy execution: (1) load-time verification catches memory-safety and termination bugs that cause the crashes described above; (2) typed maps provide concurrent state sharing between profiler and tuner without ad hoc shared memory; and (3) atomic program replacement enables verified hot-reload.

3 DESIGN

3.1 Design Tensions

T1: Safety vs. overhead. NCCLbpf verifies only at load time (1–5 ms, amortized over the job lifetime), relying on the invariant that verified BPF bytecode, once JIT-compiled, cannot violate its safety guarantees at runtime. The tradeoff: verification catches memory safety and termination bugs, not semantic correctness.

T2: Structured sharing vs. flexibility. All cross-component state sharing uses typed eBPF maps with atomic access semantics. Maps constrain policies to fixed-size keys/values, but NCCL policy state is inherently simple (per-communicator scalars), and structured sharing eliminates the locking bugs that plague ad hoc native plugin state sharing.

T3: Availability vs. consistency during updates. Hot-reload atomically swaps the function pointer; any in-progress call completes under the old policy, the next call uses the new one. No call is lost. If the new policy fails verification, the old policy continues. The tradeoff: different threads may briefly execute different policies, which is acceptable because collective decisions are independent across calls.

T4: Compatibility vs. capability. NCCLbpf operates within NCCL's existing plugin ABI, requiring no source modifications, no custom builds, and no kernel modules. The action space is limited to algorithm, protocol, and channel count selection. Deeper interventions are outside scope, but the exposed knobs cover the most impactful tuning decisions. NCCLbpf selects among NCCL's built-in algorithms; it does not implement custom collective algorithms.

3.2 Threat Model

Policies are written by platform operators, not untrusted tenants. The eBPF verifier (PREVAIL-based) checks memory safety, bounded execution, stack safety, and helper whitelisting. The system does *not* protect against resource exhaustion, side channels, bugs in the trusted computing base (bpftime JIT, PREVAIL verifier, host plugin), or semantically incorrect decisions. This provides defense in depth (eliminating crashes, hangs, and memory corruption), analogous to the Linux kernel's eBPF model.

3.3 Architecture

Figure 1 shows the architecture. Policy authors write restricted C compiled to BPF ELF objects. At load time, each program is verified and JIT-compiled to x86-64 code. The map subsystem provides intra-policy state and cross-plugin communication: a profiler eBPF program writes latency observations; the tuner reads them. Hot-reload verifies and JIT-compiles a replacement, then atomically swaps the function pointer via compare-and-swap.

Policy programming model. A policy takes a `policy_context` (collective type, message size, rank count) and writes algorithm, protocol, and channel count decisions. The verifier ensures policies only read input fields and write output fields. Map lookups require null checks before dereference—omitting one causes load-time rejection.

Listing 1: Profiler-to-tuner closed loop. The profiler (top) writes latency into a shared eBPF map; the tuner (bottom) reads it for adaptive channel selection. Null checks (lines 5, 14) are enforced by the verifier.

```
1  /* --- profiler eBPF program --- */
2  SEC("profiler")
3  int record_latency(struct profiler_context *ctx) {
4    __u32 key = ctx->comm_id;
5    struct latency_state *st =
6      bpf_map_lookup_elem(&latency_map, &key);
7    if (!st) return 0;
8    st->avg_latency_ns = ctx->latency_ns;
9    st->channels = ctx->n_channels;
10   return 0;
11 }
12 /* --- tuner eBPF program --- */
13 SEC("tuner")
14 int size_aware_adaptive(struct policy_context *ctx) {
15   __u32 key = ctx->comm_id;
16   struct latency_state *st =
17     bpf_map_lookup_elem(&latency_map, &key);
18   if (!st) { ctx->n_channels = 4; return 0; }
19   if (ctx->msg_size <= 32 * 1024)
20     ctx->algorithm = NCCL_ALGO_TREE;
21   else
22     ctx->algorithm = NCCL_ALGO_RING;
23   ctx->protocol = NCCL_PROTO_SIMPLE;
24   if (st->avg_latency_ns > 1000000)
25     ctx->n_channels = min(st->channels + 1, 16);
26   else
27     ctx->n_channels = st->channels;
28   return 0;
29 }
```

The two programs share `latency_map` via the eBPF map subsystem—a capability absent from NCCL's native plugin architecture, where tuner and profiler have no shared state. Null checks on map lookups (lines 5 and 14) are enforced by the verifier; omitting either causes load-time rejection.

4 IMPLEMENTATION

NCCLbpf is implemented in approximately 2,500 lines of C/C++ (plugin host) and 500 lines of restricted C (policy programs).

bpftime integration. We embed bpftime [18] as the userspace eBPF runtime. bpftime provides an LLVM-based JIT compiler, a PREVAIL-based [5] static verifier, and a map subsystem. The verifier runs in the same address space as the plugin, adding approximately 1–5 ms of one-time startup overhead amortized over the lifetime of the training job. The LLVM JIT produces optimized x86-64 code, narrowing the gap to native performance.

Plugin registration. NCCLbpf registers itself as both a tuner and profiler plugin via NCCL's `ncclTunerPlugin_v3` (or v5) and `ncclProfilerPlugin_v1` (or v6) interfaces. The tuner plugin's `getCollInfo()` function constructs a `policy_context`, invokes the JIT-compiled eBPF function, and copies the output fields into NCCL's result structure. The profiler plugin's event callbacks extract latency information and update the shared eBPF maps.

NCCL integration challenges. NCCL's tuner API uses cost arrays rather than direct algorithm IDs: the tuner sets costs to zero for

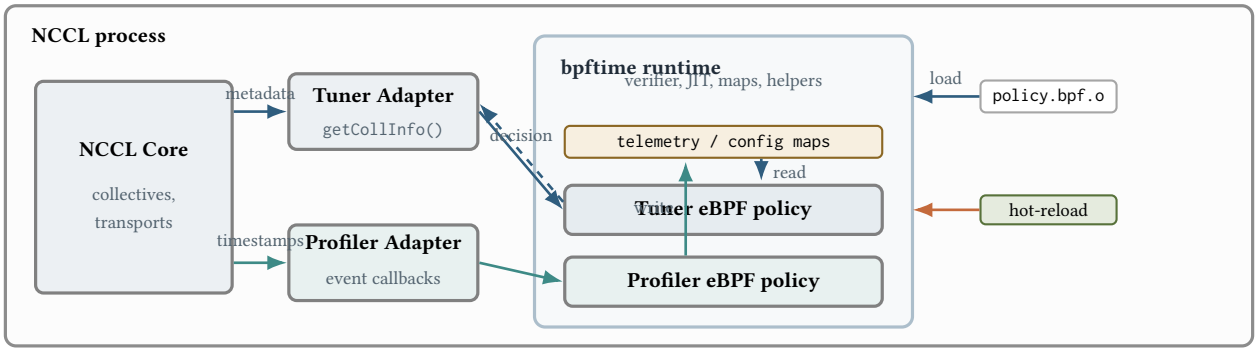

Figure 1: NCCLbpf architecture. Policy programs are verified via PREVAIL, JIT-compiled by bpftime, and execute inside the NCCL process. Profiler and tuner communicate through shared typed eBPF maps. The entire system loads as standard NCCL plugins.

preferred choices and to a sentinel value (`1e9`) for others, allowing NCCL to fall back gracefully if the requested combination is unavailable. Our integration translates eBPF policy outputs into cost array entries. NCCL also passes a maximum channel count that the tuner must respect; our native baseline layer clamps the policy's request. Communicator identification required deriving a stable ID from the context pointer via hashing, since the communicator ID is not directly exposed in the tuner API as a simple integer.

Hot-reload mechanism. The active policy is stored as an atomic function pointer. Reload has three phases: (1) verify the new BPF program, (2) JIT-compile it, (3) atomic compare-and-swap on the pointer. The total reload cost is ~7.3 ms; only the final swap (0.774 µs) touches the hot path. The old pointer is retained until in-flight calls drain. If the replacement fails verification, the swap is aborted and the old policy continues; the system never enters an unverified state.

Native baseline for comparison. To provide a fair overhead comparison, we implemented a native C++ baseline that performs identical policy logic without the eBPF layer, compiled with the same optimization flags (`-O2`). The overhead difference between the native baseline and eBPF policies directly measures the cost of the eBPF dispatch and JIT execution layers, isolated from any policy logic cost.

5 EVALUATION

Testbed. CPU microbenchmarks run on a 24-core x86-64 platform with isolated cores; each benchmark performs 1 million calls and reports P50/P99 latencies. GPU experiments use a single NVIDIA RTX 5090 with CUDA 12.9 and NCCL 2.29.7, running 2 MPI ranks on the same host over socket transport.

5.1 Overhead

Table 1 reports per-call latency. The overhead is predictable: approximately $51 + 26n_{\text{lookup}} + 25n_{\text{update}}$ ns. Even the most complex policy (`slo_enforcer`) adds only 100 ns. For a 128 MB AllReduce (175 µs), this is 0.06% of collective latency.

End-to-end GPU measurements across 10 message sizes (8 B to 128 MB) confirm that the CPU-side overhead is invisible at the GPU level: all tested configurations (no plugin, `noop`, `size_aware_v2`, `slo_enforcer`) produce statistically indistinguishable AllReduce latencies.

Table 1: CPU microbenchmark (1M calls each). eBPF policies add 52–100 ns over native code. The overhead decomposes into eBPF dispatch (+51 ns), map lookup (+26 ns each), and map update (+25 ns each).

Policy	P50 (ns)	P99 (ns)	ΔP50
Native baseline	10	16	—
`noop`	61	74	+51
`size_aware_v2`	62	73	+52
`lookup_only`	88	104	+78
`lookup_update`	113	134	+103
`adaptive_channels`	112	136	+102
`slo_enforcer`	110	134	+100

5.2 Safety and Hot-Reload

We tested 14 eBPF programs against the PREVAIL-based verifier: 7 safe policies (including all policies in Table 1) were accepted; 7 unsafe programs, each targeting a distinct bug class (null-pointer dereference, out-of-bounds access, illegal helper, stack overflow, unbounded loop, input-field write, division by zero), were rejected at load time with actionable error messages.

The safety difference is concrete. We implemented the same null-dereference bug as both a native C plugin and an eBPF policy:

```
Native plugin: Signal: SIGSEGV (address 0x0)
                in getCollInfo() at native_bad_plugin.so
eBPF policy:   VERIFIER REJECT: R0 is a pointer to
                map_value_or_null; must check != NULL
                before dereference at insn 7
```

The native plugin crashes the training job; the eBPF policy is caught before execution.

Hot-reload swaps the active policy atomically via compare-and-swap (0.774 µs swap time; ~7.3 ms including verification and JIT). Across 400,000 continuous invocations, we observe zero lost calls. If the replacement fails verification, the old policy continues uninterrupted.

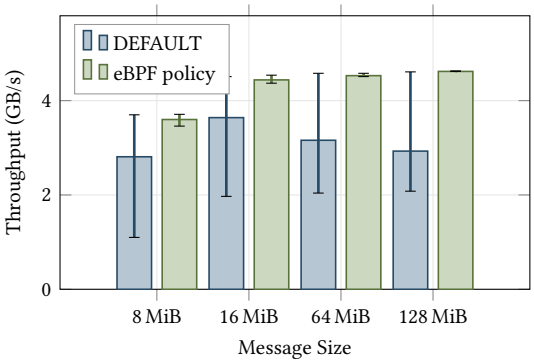

Figure 2: AllGather throughput stability (bars: mean of 3 runs; whiskers: min–max range). DEFAULT exhibits large run-to-run variance at all sizes; our eBPF policy eliminates it. RTX 5090, 2 ranks, socket transport.

5.3 Stability Improvement

NCCL's default tuner selects algorithm and protocol based on a static cost model computed at initialization. AllGather under the default configuration exhibits *high run-to-run variance*: across 3 independent runs at 128 MiB, 2 runs produced only 2.08–2.11 GB/s while the third achieved 4.61 GB/s (Figure 2). The slow runs show degraded throughput at *all* message sizes (even 1 MiB: 0.23 vs. 0.48 GB/s). We hypothesize that NCCL's topology-dependent initialization intermittently selects a suboptimal internal protocol configuration.

Forcing `NCCL_PROTO=Simple` eliminates this variance entirely: all 3 runs produce 4.62±0.01 GB/s at 128 MiB. When DEFAULT operates normally (run 3), its throughput matches Simple (4.61 vs. 4.63 GB/s), suggesting that the degradation is not inherent to the protocol choice but to non-deterministic initialization.

Our `size_aware_adaptive` policy (Listing 1) always selects Simple protocol, which, as a side effect, eliminates this run-to-run variance. This demonstrates a policy value beyond throughput optimization: *improved stability* through consistent protocol steering. The policy adds zero measurable GPU-level overhead versus NCCL's default when DEFAULT operates in its fast mode.

6 RELATED WORK

Collective communication. MSCCL [3] and TACCL [14] synthesize static schedules offline; they do not address runtime policy selection or safety. AutoCCL [2] automates tuning via search but uses native code without verification. MSCCL++ [13] provides GPU-driven primitives at the mechanism layer. Hu et al. [6] provide the first comprehensive analysis of NCCL's protocol variants and algorithms. NCCLbpf operates at the policy layer and is complementary: a policy can activate synthesized schedules based on dynamic conditions.

eBPF for systems extensibility. gpu_ext [20] applies eBPF to GPU driver-level resource management (scheduling, memory placement); NCCLbpf targets the collective communication library layer, where policies govern algorithm and protocol selection rather than device resources. XRP [21] embeds eBPF in the kernel's NVMe path, exploiting the same insight that certain system paths are latency-tolerant enough for eBPF overhead. Electrode [22] accelerates distributed protocols; eTran [8] applies eBPF to RDMA network functions; PageFlex [17] extends eBPF to page management. bpftime [18] provides the userspace eBPF runtime substrate that NCCLbpf builds upon.

Alternative safe-extension mechanisms. WebAssembly provides sandboxed execution but lacks eBPF's static termination and memory-safety verification; its runtime overhead is typically higher due to bounds checks that eBPF eliminates statically. Out-of-process services provide strong isolation but add microseconds of IPC latency, which is an order of magnitude more than NCCLbpf's 52–100 ns in-process overhead. Declarative DSLs cannot express stateful policies or feedback loops. Hardware isolation (Intel MPK) provides memory separation without verification. eBPF provides both verification and efficient execution in a single model.

7 DISCUSSION

This paper focuses on NCCL's tuner and profiler plugin interfaces; extending coverage to the net and env plugins is straightforward future work. Our evaluation uses 2 ranks on a single node over socket transport; multi-node experiments with InfiniBand and large rank counts are needed to validate the approach at production scale. A higher-level policy DSL compiled to BPF would lower the barrier for ML engineers who are not familiar with eBPF programming.

The eBPF-for-extensions pattern may generalize beyond NCCL. AMD's RCCL [1] shares the same plugin architecture, making cross-vendor portability feasible. Other libraries with native-code extension points (MPI implementations, RDMA middleware) face similar safety gaps, and the observation that GPU-adjacent CPU-side decision paths have generous latency budgets for verified execution may apply broadly across the accelerator software stack.

8 CONCLUSION

We presented NCCLbpf, a system that transforms NCCL's unsafe native plugin extension points into verified, composable policy hooks by embedding a userspace eBPF runtime. The key insight is that NCCL's CPU-side decision path is latency-tolerant enough to absorb eBPF overhead. NCCLbpf adds 52–100 ns per decision (under 0.06% of collective latency), catches crash-inducing bugs at load time, supports atomic hot-reload (~7.3 ms total, 0.774 µs on the hot path) without call loss, and reduces AllGather throughput variance from 50% to 0.2% coefficient of variation on our testbed, with zero NCCL modifications.

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